# Computer Organization

CodeHS

# What are the main parts of a computer?

# Two Main Categories

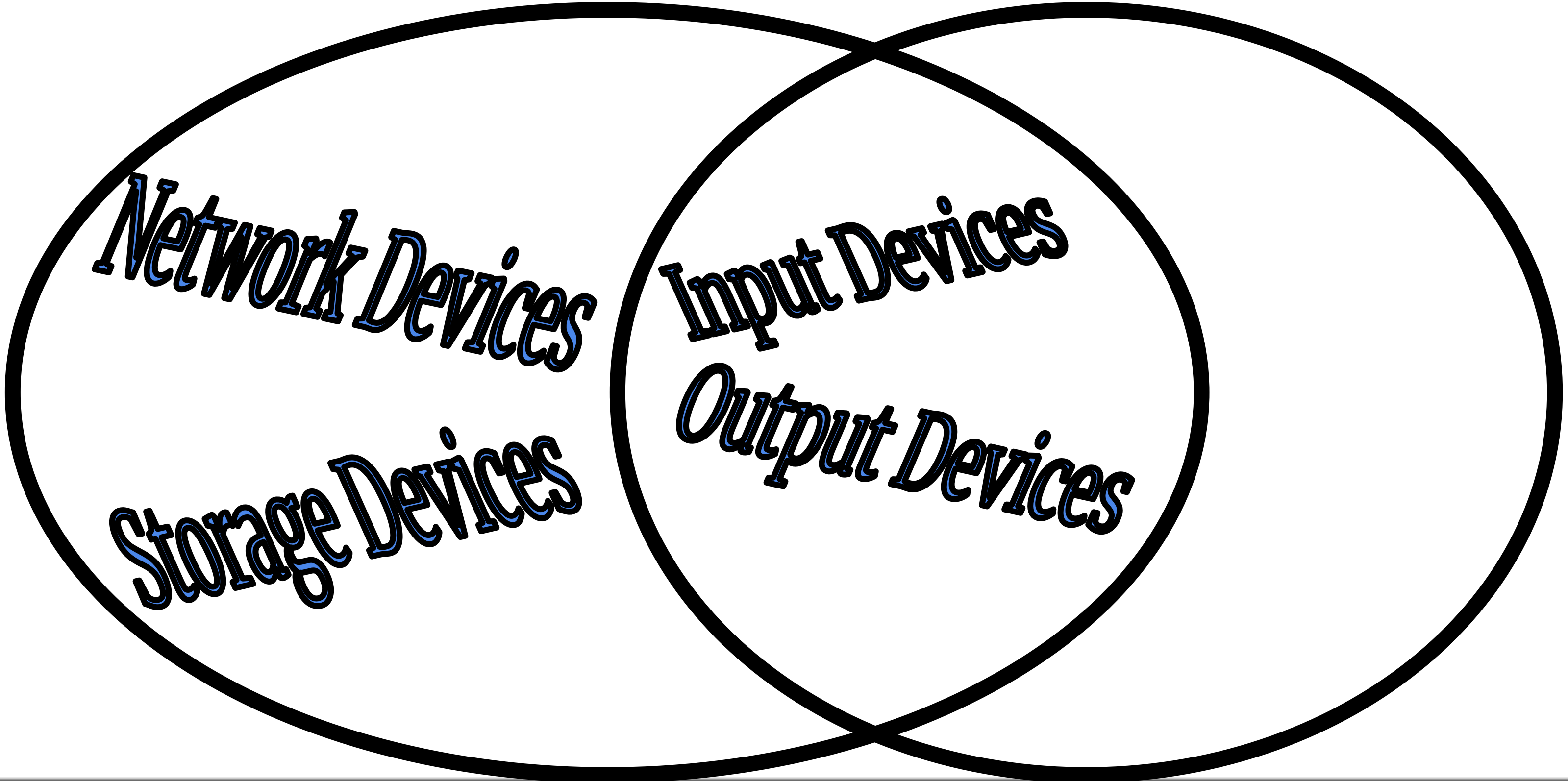
Computers parts divided into

1. Hardware
   - Physical components

2. Software
   - Programs that run on the computer

# Several Sub-Categories

Hardware




Software

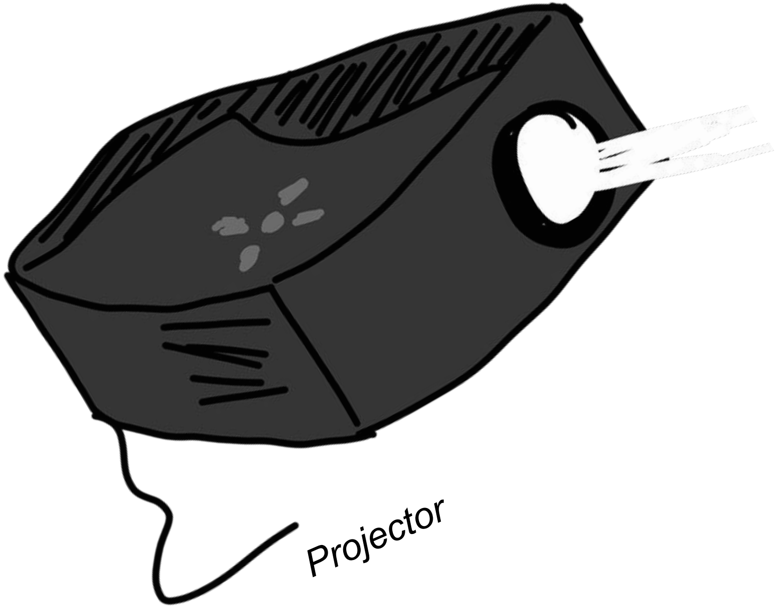
# Output Devices

How your computer
tells you things.

# Output Devices

- Allows your computer to
  - Display information
  - Give audio signals
  - Transform digital information into physical information

# Output Devices That Display



Projector

- Monitors
- Video Card
- Projector

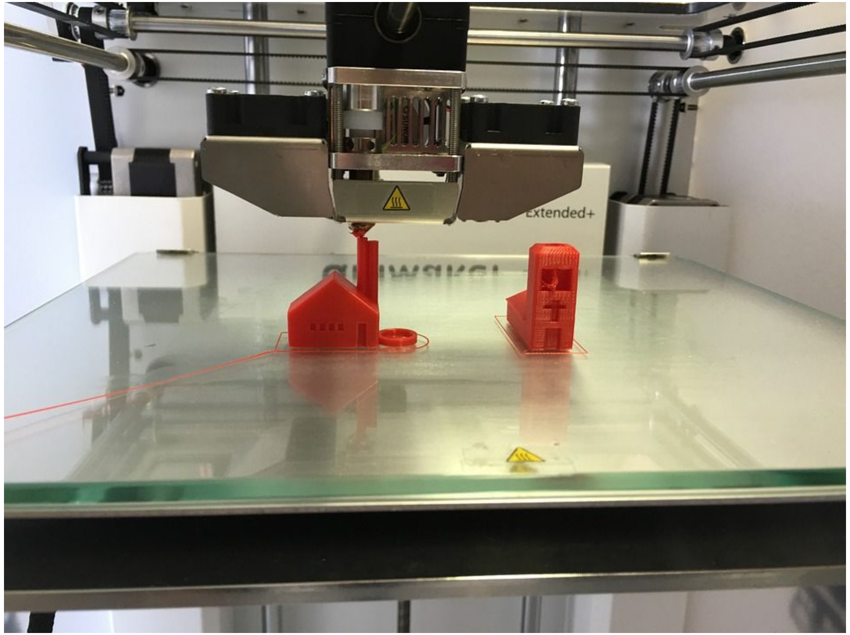
Monitor

# Output Devices for Audio

- Speakers
- Sound Card
- Headphones

# Output Devices for Digital to Physical Conversion

- Laser or Inkjet printer
- 3D printer




*3D Printer*

# Input Devices

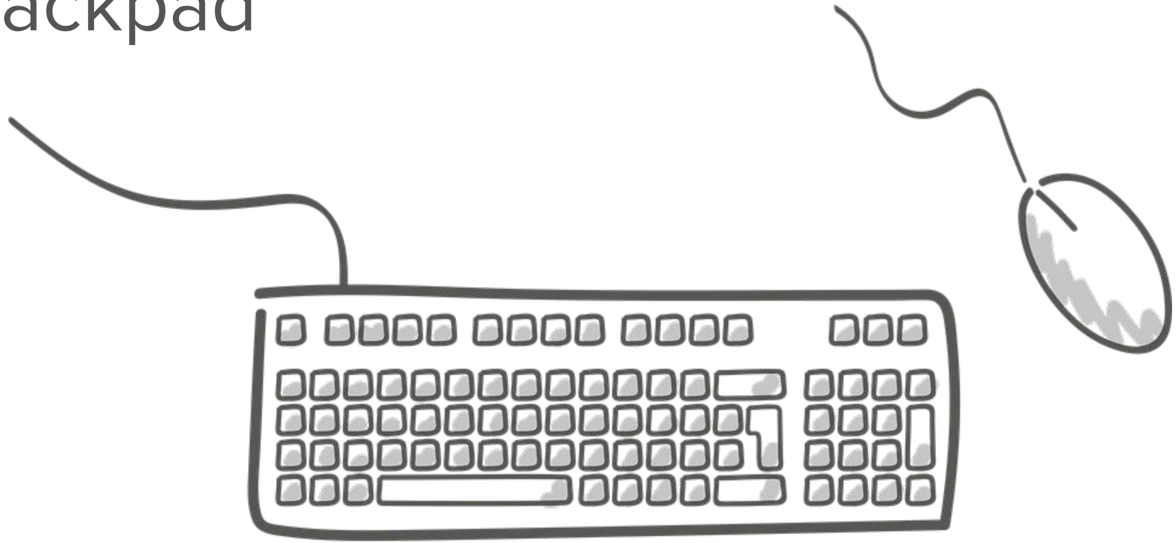
How you give information to your computer.

# Input Devices

- Allow you to
  - Interact with the computer
  - Turn physical data into digital data

# Interactive Input Devices

- Touchpad or Trackpad
- Mouse
- Keyboard
- Joystick
- Remote
- Touchscreen



# Digitizing Input Devices

- Scanner
- Microphone
- Digital Cameras
- Webcams
- Sensors


*Airplane cockpit. Airplane sensors gather information like altitude and direction. The dials are output devices, but they get information the input devices -- the sensors!*

# Storage Devices

How your computer remembers information

# Storage Devices

- Computer's memory
  - Allows computer to store information to retrieve later
  - Allows users to save data
- Two Categories
  - Primary storage
  - Secondary storage

# Primary Storage

- Memory the computer uses to perform its normal activity
- Memory is cleared when power is turned off
- Called RAM
    - Random Access Memory
    - More info later

# Secondary Storage

- Extra memory used for persistent storage
- Examples
  - CD
  - USB drive
  - Hard drive
  - Floppy disk
  - Cloud storage

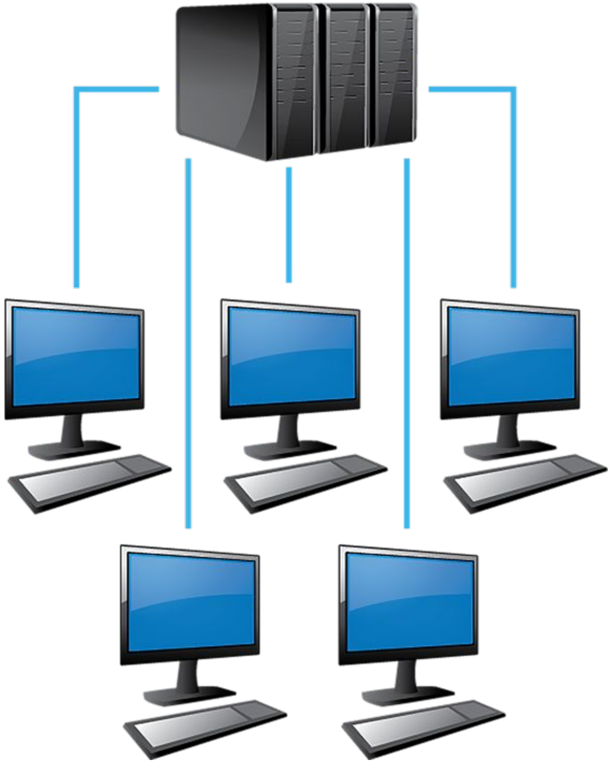
*Floppy Disk*

# Network Devices

How your computer communicates with other computers.

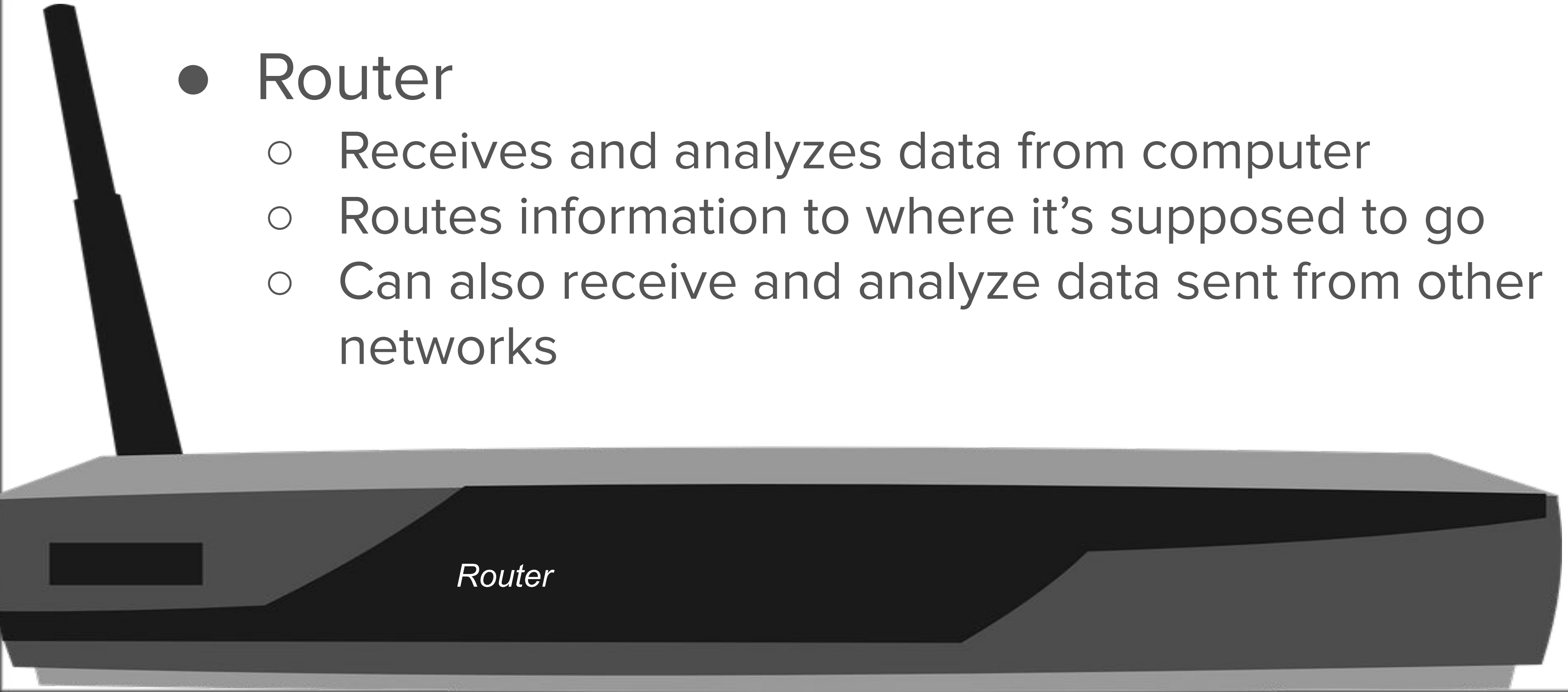
# Networks

- Computers usually don’t directly communicate
- Sends data to a central location, which then passes on the information



# Router: Center of the Network

- Router
  - Receives and analyzes data from computer
  - Routes information to where it's supposed to go
  - Can also receive and analyze data sent from other networks



# Local Area Networks

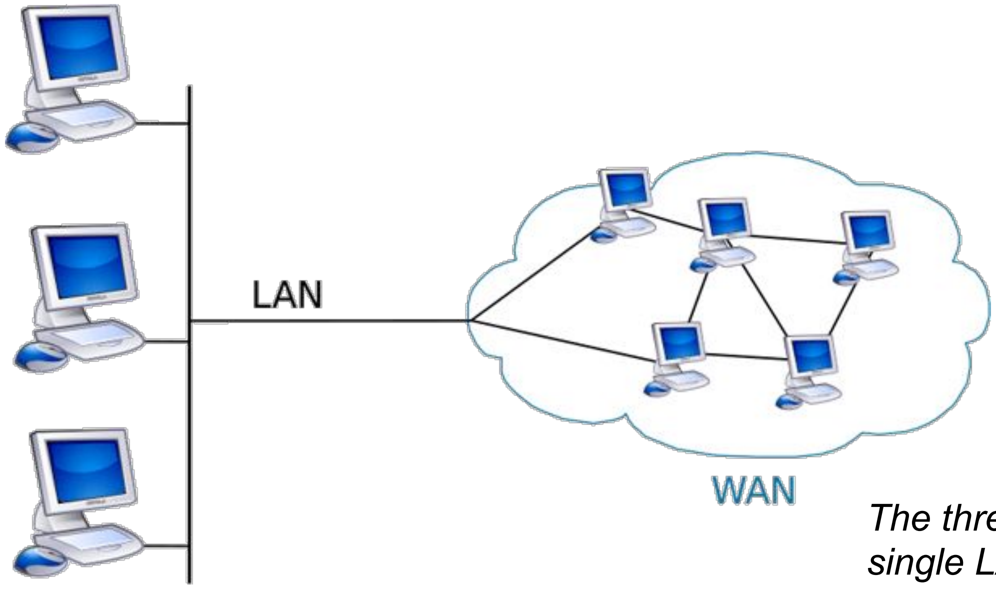
- LAN for short
- Connects computers close to each other
  - e.g. In the same building
- Can usually be connected with a single router

# Wide Area Networks

- WAN for short
- Connects computers far away from each other
- Needs multiple routers and special cables

# LAN vs WAN

- A WAN is usually made up of several LANs




*The three computers on the left make up a single LAN. The LAN is part of the larger WAN, which connects several LANs*[1]

# Network Devices

- Router
  - Receives and analyzes data sent from other networks
- Ethernet port
  - Allows an ethernet cable to be attached
  - Ethernet is one way to physically connect to a network
- Wireless Internet Card (WIFI Card)
  - Allows a computer to connect to the internet

# Sources


Pictures

- https://pixabay.com/
- [1] "LAN vs WAN." Diffen.com. Diffen LLC, n.d. Web. 14 Jul 2016.< http://www.diffen.com/difference/LAN_vs_WAN >

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